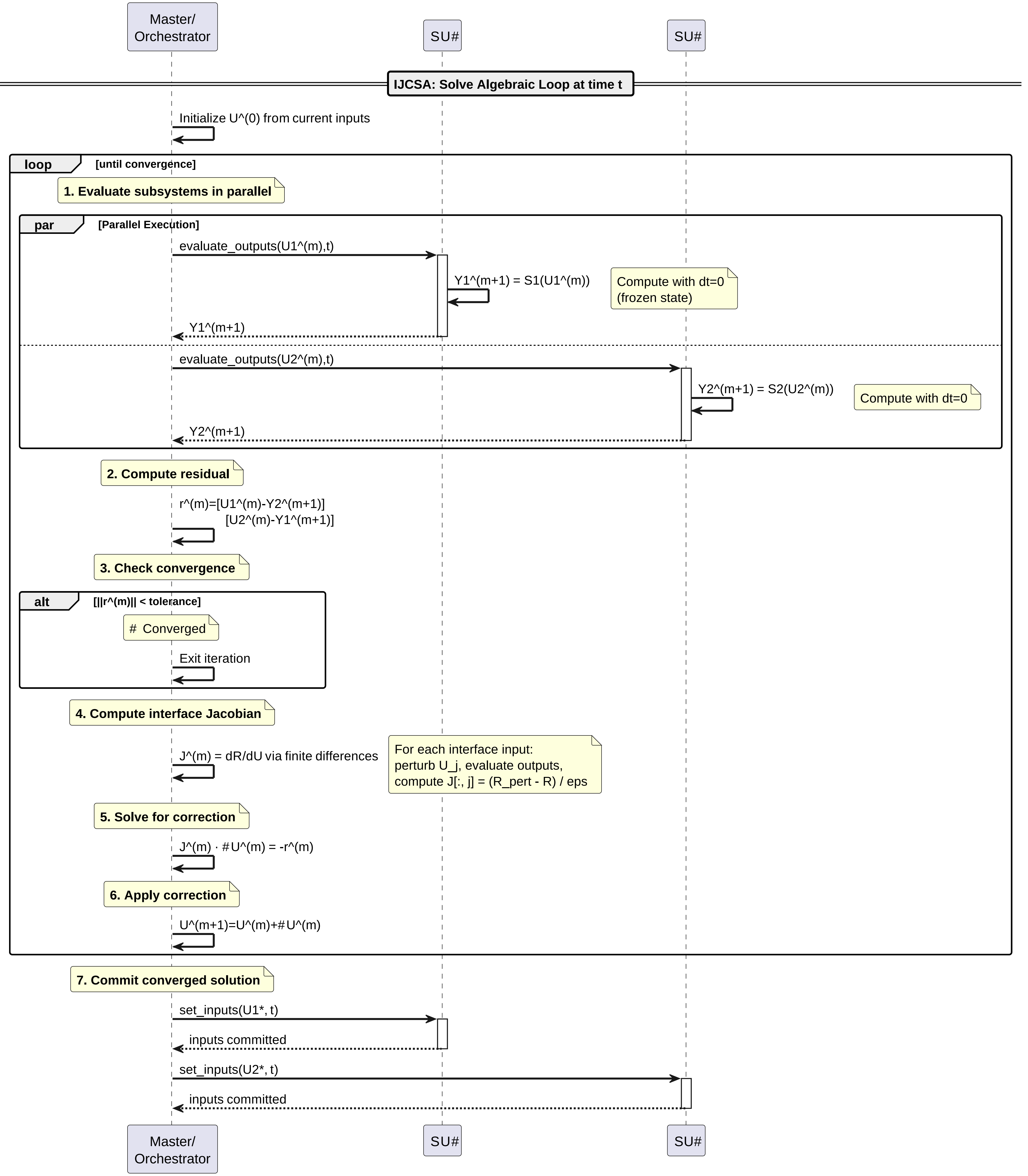
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IJCSA: Solve Algebraic Loop at time t
Initialize U^(0) from current inputs
loop [until convergence]
1. Evaluate subsystems in parallel
par [Parallel Execution]
evaluate_outputs(U1^(m),t)
Y1^(m+1) = S1(U1^(m))
Compute with dt=0 (frozen state)
Y1^(m+1)
evaluate_outputs(U2^(m),t)
Y2^(m+1) = S2(U2^(m))
Compute with dt=0
Y2^(m+1)
2. Compute residual
r^(m)=[U1^(m)-Y2^(m+1)] [U2^(m)-Y1^(m+1)]
3. Check convergence
alt [||r^(m)|| < tolerance]
Converged
Exit iteration
4. Compute interface Jacobian
J^(m) = dR/dU via finite differences
For each interface input: perturb U_j, evaluate outputs, compute J[:, j] = (R_pert - R) / eps
5. Solve for correction
J^(m) · #U^(m) = -r^(m)
6. Apply correction
U^(m+1)=U^(m)+#U^(m)
7. Commit converged solution
set_inputs(U1*, t)
inputs committed
set_inputs(U2*, t)
inputs committed
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